

Prasanna Sand

sandprasannap@gmail.com | +1 (323) 812 – 8857 | San Jose, CA | [LinkedIn: prasannasand](https://www.linkedin.com/in/prasannasand) | [Devpost: prasannasand12](https://devpost.org/prasannasand12) | prasannasand.com

Education

- Master of Science in Computer Science and Engineering | Santa Clara University** **09/2023 – 06/2025**
Courses: Distributed Systems, Algorithms, Advanced Operating Systems, Advance Computer Architecture, Object Oriented Analysis Design and Programming, Pattern Recognition and Data Mining, Machine Learning, Computer Networks, Solutions Architecture and the Cloud.
- Master of Business Administration | Savitribai Phule Pune University | Pune, India** **07/2021 – 08/2023**
Courses: Strategic Management, Global Strategic Management, Decision Science, Business Research Methods, Digital Business, Enterprise Performance Management, Behavioral Finance, Business Valuation, Corporate Financial Restructuring, Strategic Cost Management.
- B. Tech in Computer Science and Engineering | MIT World Peace University | Pune, India** **07/2018 – 05/2021**
Courses: Design and analysis of algorithms, Database Systems, Operating Systems, Computer Architecture, High Performance Computing, Artificial Intelligence, Data warehouse and mining, System Software and Compilers, Computer Vision, Natural Language Processing.

Skills

- Languages:** Java, C, C++, JavaScript, Typescript, Python, HTML, CSS, SQL, Swift, Bash.
- Cloud & DevOps:** AWS, Docker, Git, JIRA, Postman, Google Cloud, Kubernetes, Firebase, CI/CD (GitHub Actions).
- Libraries and Framework:** React, Angular, Node.js, TensorFlow, Keras, OpenCV, Flask, Express, Spring Boot, Flutter.
- Database:** MySQL, DuckDB, MongoDB (NoSQL).
- Core competencies:** Full Stack Development, Design Patterns, Unit Testing, Microservices, Distributed Systems, High-Concurrency Optimization, REST APIs, Agile, AI/ML Model Integration (LLMs, Gemini API), Database Systems, Storage Engines, Query Optimization.

Experience

- Software Engineer | Frugal Innovation Hub | Santa Clara University | Santa Clara, California** **12/2024 – Present**
- Delivered production-ready cross-platform applications in JavaScript, Node.js, and Firebase, used by non-profit partners to serve real users.
 - Designed modular architecture (presentation, domain, data layers), ensured 85%+ unit test coverage and enforced code reviews for maintainability.
 - Automated CI/CD pipelines with GitHub Actions, supporting multi-platform builds and streamlined QA workflows, increasing deployment speed by 40%.
 - Partnered with engineers, analysts, and external stakeholders, ensuring solutions met production quality, scalability, and maintainability standards.
 - Facilitated 30+ peer code reviews and resolved 50+ live bugs pre-production, reducing QA-reported issues by ~25% while authoring technical documentation to improve team-wide knowledge sharing.
- Computer Vision Intern | NextLeap Aeronautics | Bangalore, India** **06/2020 – 08/2020**
- Engineered a collision-prevention system using YOLOv3 in C++, achieving 92%+ mAP accuracy at 30 FPS in UAV test flights, enhancing real-time obstacle avoidance.
 - Optimized power line detection pipeline (PCNF Filtering, Gabor Filter, Hough Transform) with algorithmic refinements, boosting detection precision by ~35% while reducing false positives.
 - Prototyped high-speed stereo-vision algorithms (Pushbroom Stereo) capable of processing 60+ FPS, enabling UAV navigation at 20m/s with minimal latency.
 - Explored multi-sensor fusion (vision + IMU + LiDAR) to reduce localization drift by ~20%, improving system reliability in GPS-denied environments.

Projects

- Distributed Log Storage & Query System (DLQS)** **04/2025 – Present**
- Prototyped a production-scale distributed log-structured storage system in C++ with Java bindings, implementing replication, leader election, and compaction for fault-tolerant, high-throughput persistence.
 - Integrated vectorized execution and DuckDB embedding to support SQL-like queries, reducing query latency by ~30% under 100M+ record workloads.
 - Applied profiling and debugging tools (gdb, Valgrind, perf) to identify bottlenecks and improve system health and reliability.
- BrowseAble – AI Accessibility Chrome Extension** **04/2025**
- Developed a Chrome extension using React and Gemini API to adapt web pages for neurodivergent users (ADHD, Autism, SPD).
 - Implemented user-role authentication (user vs. caretaker) and integrated Firebase for personalized accessibility features.
 - Designed dynamic content overlays and simplified layouts using AI-generated DOM adaptation for inclusive browsing.
 - Enabled real-time summarization, font adjustments, visual simplification, and caregiver dashboards, supporting 1,000+ web sessions in testing with <200ms latency.
- Distributed and Hybrid Query Engine for DuckDB** **11/2024 – Present**
- Extended DuckDB's C++ codebase with a distributed query execution layer, enabling hybrid ETL workflows across heterogeneous data sources.
 - Designed and implemented low-level query optimization + workload distribution algorithms in C++, improving latency by ~25% on multi-node clusters.
 - Integrated C++ concurrency primitives and scheduling to handle query parallelism and fault tolerance at scale.
 - Added system health monitoring + performance debugging (gdb, Valgrind, perf) to ensure cluster reliability under 100M+ row workloads.
- AllBank – AI-Driven Inclusive Banking Solution** **11/2024**
- Developed a holistic financial management app using React Native and Firebase, offering traditional banking services, goal-based savings, QR code payments, and AI-powered financial insights to empower individuals and small businesses.
 - Integrated Google Gemini AI and NLP technologies to deliver real-time financial assistance, personalized analytics, and human-like conversational support, enhancing accessibility and decision-making for underbanked populations.
- Online Ordering App, DevC 2.0 Hackathon** **06/2024 – 07/2024**
- Developed an AI-powered food ordering app with personalized meal recommendations based on user health data, dietary restrictions, and calorie goals, using Express.js for backend architecture and React.js for a responsive user interface.
 - Leveraged MongoDB for efficient data storage and Node.js to handle server-side logic, ensuring seamless data flow and user interactions.
 - Used Docker to containerize and deploy the app, optimizing performance and scalability across multiple environments.

Publications

- Brainwave Controlled Wheelchair- Vol-07, Issue-02. ([IJREAM](#))
- Applications of computer vision in Agriculture- 2021. ([IJEAST](#))

Certificates and Achievements

Hackathons – **Winner** (HackUTD 2024 - Goldman Sachs Track), **Finalist** (TikTok TechJam Hackathon 2024, ByteDance Ltd.) | AWS Certified Solutions Architect – Associate | Certified Business Analytics | DeepLearning.AI TensorFlow Developer | Certified Google Cloud Engineer | **Co-Founder**, Well-Whizer (Wearable technology to promote AI for Mental Health and track client's essential information to assist the counselors)